

Measuring Semantic Redundancy in Multilingual Training Corpora

Complete experimental results — v0 through v4

Author: K7007 | GitHub: github.com/kesavanvjp0007/project-uctf | July 2026 | Pre-print

TOTAL PAIRS	LANGUAGES	ENCODERS	BEST MEAN SIM	SENT/SEC
13,320	7	4	0.882	1,515
all experiments	EN ZH JA RU NE SI KY	MiniLM mE5 LaBSE mp	mE5-small v4	RTX 3050 GPU

1 Research question

How much semantic duplication actually exists across languages in real multilingual AI training datasets — and can embedding-based methods reliably detect it?

This paper investigates whether cross-lingual semantic redundancy is measurable, substantial, and encoder-dependent. The findings inform the feasibility of the Universal Compressed Training Format (UCTF) hypothesis: that a machine-native compressed representation could replace raw multilingual text as an AI training format.

2 Experimental progression — v0 to v4

v0 Controlled (20 pairs)		0.9570
v1 Real corpus (1,500 pairs)		0.7470
v2 mE5 encoder (1,500 pairs)		0.8820
v3a Noise robustness (900)		0.9630
v3b Low resource (600 pairs)		0.9050
v4 Corpus scale (10,000)		0.8950

3 Encoder comparison — v2 (1,500 pairs, GPU)

Encoder	Eng-Chinese	Eng-Japanese	Eng-Russian	Overall mean	Neg. pairs	Std dev
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MiniLM	0.7948	0.6589	0.7886	0.7474	1	0.1878
mE5	0.8694	0.8630	0.9136	0.8820	0	0.0455
LaBSE	0.7898	0.6149	0.7855	0.7301	0	0.1990
mpnet	0.8249	0.7044	0.8176	0.7823	0	0.1733

mE5-small (row highlighted) outperforms all tested encoders — highest mean (0.8820), lowest standard deviation (0.0455), zero negative similarity pairs, and near-elimination of the English-Japanese divergence gap observed with MiniLM (0.136 → 0.006).

4 Noise robustness — v3a (mE5, 300 sentences per type)



mE5 maintains 94.9–97.5% similarity to clean text across all noise types. All noise variants scored above the 0.85 threshold in 98.7–100% of cases, confirming viability as a UCTF encoding baseline for real-world noisy corpora.

5 High resource vs low resource — v4 corpus scale (2,000 pairs each)



Counterintuitively, low-resource languages (Sinhala 91.8%, Nepali 90.4%) outperformed high-resource languages (Chinese 87.6%, Japanese 86.7%) at corpus scale. This unexpected finding is flagged for investigation in Paper 2.

6 Key findings

Text deduplication fails completely

Exact duplicate detection finds 0% cross-lingual redundancy. Semantic embedding approaches are necessary — not optional.

mE5-small is the recommended baseline

Mean 0.882, std 0.046, zero negative pairs. Closes Eng-Jap gap from 0.136 (MiniLM) to 0.006.

mE5 is robust to real-world noise

94.9–97.5% similarity to clean text under character errors, word shuffling, and code-switching.

Low resource matches high resource

Sinhala (91.8%) and Nepali (90.4%) outperform Chinese (87.6%) and Japanese (86.7%) at corpus scale.

Consumer GPU is sufficient

20,000 sentences encoded in 13.2 seconds at 1,515 sent/sec using only 458 MB VRAM on RTX 3050.

7 Complete experimental log

Version	Scale	Dataset / Setup	Mean sim	Key finding
v0	20 pairs	Manual aligned, 4 languages, MiniLM	0.9570	100% precision at all thresholds
v1	1,500 pairs	OPUS-100 real corpus, 3 lang pairs, mE5	0.7417	Eng-Jap gap 0.136 identified
v2	6,000 rec.	4 encoders compared on GPU	0.8820	mE5 best — zero negatives, std 0.046
v3a	900 noisy	3 noise types (char/shuffle/switch), mE5	0.9627	98.7–100% above 0.85 threshold
v3b	600 pairs	Nepali, Sinhala, Kyrgyz low-resource	0.9045	Low resource matches high resource
v4	10,000 pairs	5 languages, corpus scale, GPU RTX 3050	0.8948	1,515 sent/sec · 458 MB VRAM used

8 Limitations of Paper 1

- OPUS-100 is a parallel translation corpus — cleaner than real web-crawled training data.
- Only 3 high-resource and 3 low-resource language pairs tested — broader coverage needed.
- Single noise simulation per type — real corpus noise is more varied and compound.
- mE5-small tested, not mE5-large or SONAR — stronger encoders may show different characteristics.
- No manual audit of false positives completed — automated pipeline only at this stage.
- Negative similarity cases (-0.044) require manual inspection to determine root cause.

9 Conclusion

Cross-lingual semantic redundancy is measurable, substantial, and encoder-dependent. mE5-small is a viable baseline for a UCTF compression layer — robust to noise, effective across language families including low-resource languages, and practical on consumer hardware. The core hypothesis of Project UCTF — that a machine-native compressed representation could reduce multilingual training data redundancy — is supported by these preliminary findings.

"Cross-lingual semantic redundancy is measurable, substantial, and encoder-dependent. mE5-small is a viable baseline for a UCTF compression layer — robust to noise, effective across language families, and practical on consumer hardware."



Next steps — Paper 2

- Characterize which information is universal vs language-specific using UCTF-encoded pairs.
- Manual audit of candidate_pairs.csv — label false positives and borderline cases.
- Add SONAR and LaBSE to encoder comparison at corpus scale.
- Test on web-crawled corpora (FineWeb2, CulturaX) for realistic training data behavior.
- Investigate the low-resource language outperformance finding from v4.

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